

TASNIM JAHAN

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[GitHub](#) — [Google Scholar](#) — [LinkedIn](#) — [Website](#)

Research Interests

• Computer Vision • Representation Learning • Few-Shot Learning • Transfer Learning • Domain Adaptation

Education

M.Sc. in Computer Science and Engineering (Data Science), CGPA: 3.92 / 4.00 Oct 2021 – Jun 2024
United International University Dhaka, Bangladesh
Summa Cum Laude (Highest Distinction), CGPA: 4.00 / 4.00 in last 5 semesters

M.Sc. in Mathematics, CGPA: 3.77 / 4.00 Jan 2016 – Apr 2017
Comilla University Cumilla, Bangladesh
Thesis Title: Bernoulli's equation with the variation of velocity, density, and diameter of pipe
Advisor: Zillur Rahman

B.Sc. in Mathematics, CGPA: 3.61 / 4.00 Mar 2010 – Jan 2016
Comilla University Cumilla, Bangladesh
Project Title: Numerical study for solving initial value problems in ordinary differential equations using single-step and multi-step methods
Advisor: Prof. Dr. Khalifa Mohammad Helal

Research Experience

Independent Researcher (Computer Vision & Deep Learning) Jan 2024 – Present
Remote

Advisor: Prof. Andrew King (King's College London, UK); Prof. Swakkhar Shatabda (BRAC University, Bangladesh)

- Conducting research on few-shot learning, visual representation learning, and cross-domain adaptation with a focus on robust model generalization.
- Designed and evaluated deep learning pipelines using CNNs, Vision Transformers, and pretrained vision foundation models for robust visual learning.
- Developed data preprocessing and augmentation pipelines to improve model robustness across datasets.
- Performed experimental analysis including model comparison, ablation studies, and cross-domain evaluation.
- First author of a Q1 journal manuscript (under revision) on a lightweight attention-enhanced segmentation framework.
- Co-author of an IEEE SMC 2025 paper on efficient deepfake detection using sparse subnetworks.

Publications

- L. Al Amin, Md. Ismail Hossain, Thanh Thi Nguyen, **Tasnim Jahan**, Mahbubul Islam, Faisal Quader. **Uncovering Critical Features for Deepfake Detection through the Lottery Ticket Hypothesis**. *IEEE SMC*, 2025.
- **Tasnim Jahan**, Md Easin Arafat, Swakkhar Shatabda. **ThyCLIPNet: A BiomedCLIP-Guided Lightweight Attention-Enhanced DeepLabV3+ Framework for Robust Thyroid Nodule Segmentation**. Under revision (Q1 journal).
- **Tasnim Jahan**, Andrew King, Swakkhar Shatabda. **HyProDINO: A Hybrid Prototype-guided DINO Framework for Cross-Domain Few-Shot Medical Image Segmentation**. Submitted to the Medical Imaging with Machine Learning (MIML) Workshop, MICCAI 2026.

Awards & Scholarships

Merit-based Scholarship (5 times) at United International University for academic performance Jan 2023 – Aug 2024

Ranked 5th in LISA 2025 Challenge, RISE-MICCAI Summer School May 2025 – Oct 2025

Mentee in RISE-MICCAI Mentorship Program Oct 2025 – Oct 2026

Projects

Hippocampus Segmentation (Ultra-low-field MRI) | *Python, MONAI, PyTorch* | [Code](#) Jul 2025 – Sep 2025

- Developed a MONAI-based 3D deep learning pipeline for bilateral hippocampus segmentation from ultra-low-field MRI.
- Implemented and compared Attention U-Net, DynUNet, SegResNet, and UNETR using class-balanced cropping and sliding-window inference.
- Among the tested models, Attention U-Net achieved the highest validation Dice (≈ 0.60) with post-processing.

Thyroid Nodule Segmentation | *Python, TensorFlow/Keras* | [Code](#) — [Report](#) Nov 2023 – Feb 2024

- Developed an end-to-end deep learning pipeline for thyroid nodule segmentation on the TN3K ultrasound dataset.
- Implemented and compared U-Net variants with pretrained encoders (VGG16, VGG19, ResNet50, MobileNetV2).
- Best among the tested encoders: MobileNetV2-based U-Net (Dice: 0.847, IoU: 0.738), achieving strong accuracy with a lightweight backbone.

Spatio-Temporal Data Analysis Using Deep Learning (Survey Paper) | *Literature Review* | [Report](#) Jul 2023 – Nov 2023

- Conducted a structured survey of deep learning methods for spatio-temporal data: data types, pipelines, and models.
- Covered CNNs, RNNs, GNNs, and Transformers for prediction, classification, clustering, and anomaly detection.
- Summarized real-world applications, open challenges, and research directions.

Hypo-thyroid Disease Prediction | *Python, Scikit-learn* | [Code](#) — [Report](#) Feb 2022 – May 2022

- Developed a machine learning pipeline for early detection of hypothyroid disorder using clinical and demographic data.
- Applied preprocessing techniques including imputation, feature encoding, scaling, and SMOTE for class imbalance.
- Trained and compared Decision Tree, Logistic Regression, SVM, and Random Forest classifiers.
- Achieved the best performance with Random Forest (Accuracy: 98.7%), outperforming other models.

Face Mask Detection | *Python, TensorFlow, OpenCV* | [Code](#) — [Report](#) Mar 2023 – Jul 2023

- Implemented a full pipeline for mask vs. no-mask detection using a MobileNetV2-based CNN.
- Included training, evaluation, and real-time inference with OpenCV.

Predict Customer's Future Purchases | *Python, Machine Learning* | [Code](#) — [Report](#) Jun 2022 – Sep 2022

- Developed a machine learning pipeline to predict customer purchases within 90 days using transactional retail data.
- Trained and evaluated Logistic Regression, Decision Tree, Random Forest, and Gaussian Naïve Bayes.
- Achieved the best performance with Logistic Regression (Accuracy: 99.1%, F1-score: 98.5%), while GaussianNB provided the fastest inference.

Bangla Financial Sentiment Classification | *Python, ML* | [Code](#) — [Report](#) Jan 2023 – Aug 2023

- Collected and labelled 1.8K+ Bangla financial headlines from the Prothom Alo newspaper for sentiment classification.
- Achieved the best performance with SVM (F1: 78%) and Logistic Regression (Accuracy: 76.6%).

LLM-Enhanced Stock Movement Prediction | *ML* | [Report](#) Aug 2023 – Sep 2023

- Integrated financial time-series data with information extracted from news using LLMs for weekly stock prediction.
- Improved Logistic Regression (Accuracy: 54.8% $\rightarrow$ 63.0%, F1: 42.1% $\rightarrow$ 60.9%).

IoT-Based Watering Automation and Security System for Plants & Pets | *Sensors* | [Report](#) Jul 2022 – Nov 2022

- Designed an IoT concept for automated watering with basic security monitoring for plants and pets.
- Focused on automation logic, sensor-driven triggers, and system-level design.

Skills

Programming: Python, SQL, Java, C, C++

Deep Learning: PyTorch, TensorFlow, Keras, MONAI

Computer Vision / ML: OpenCV, scikit-learn, NumPy, Pandas

Research Tools: LaTeX, Kaggle, Google Colab, Jupyter Notebook

Visualization: Matplotlib, Power BI

Soft Skills: Complex problem solving, rapid skill acquisition, effective communication

Relevant Coursework

- Advanced Artificial Intelligence
- Deep Learning
- Machine Learning
- Data Mining
- Natural Language Processing
- Research Methodology
- Advanced Database Systems
- Differential & Integral Equations
- Discrete Mathematics
- Linear Algebra
- Numerical Analysis
- Ordinary & Partial Differential Equations
- Methods of Statistics
- Vector Calculus & Tensor Analysis

Teaching Experience

Section Leader (Volunteer)

Apr 2026 – May 2026

Stanford University – Code in Place

Remote

- Selected through a competitive process from 2,000 applicants to join a global teaching team.
- Taught Python programming to a diverse cohort in a large-scale online course.
- Led problem-solving sessions and mentored students to improve programming and analytical skills.

Instructor – Mathematics

Jan 2017 – Mar 2018

Institute of Computer Science and Technology (ICST)

Feni, Bangladesh

- Taught mathematics to 200+ diploma-level students across engineering disciplines.
- Designed lectures, assignments, and assessments tailored to diverse learning needs.

Junior Teacher – Mathematics

Mar 2018 – Dec 2018

Mainamati English School and College

Cumilla, Bangladesh

- Delivered mathematics instruction in English medium and monitored student performance.
- Founded and led a math club to improve student engagement.

Professional Experience

Freelance Data Analyst

Jan 2023 – Feb 2024

Fiverr

Remote

- Developed interactive dashboards and data-driven reports using Microsoft Power BI.
- Performed data cleaning, statistical analysis, and feature engineering on real-world datasets.

Field Data Collector

Nov 2021 – Dec 2021

BRAC James P Grant School of Public Health

Dhaka, Bangladesh

- Collected qualitative and quantitative data from ~100 participants using Kobo Toolbox.
- Ensured data quality, consistency, and proper documentation for analysis.

Programme Organizer – Education

Jan 2022 – Apr 2022

Jagorani Chakra Foundation

Cox's Bazar, Bangladesh

- Supported the implementation of a UNICEF-funded education program by training teachers in effective teaching methodologies and classroom practices.
- Collaborated with stakeholders, monitored field activities, and contributed to reporting and documentation.

Education Facilitator – Education

Jan 2019 – Apr 2020

DanChurch Aid

Cox's Bazar, Bangladesh

- Delivered literacy and numeracy sessions to ~300 learners in a UNICEF-funded program.
- Designed course materials and conducted baseline and endline data collection.

Certifications & Courses

WorldQuant University: Computer Vision Lab

Dec 2025 – Present

RISE-MICCAI Summer School (MICCAI Society)

Jul 2025

Stanford University: Code in Place

Apr 2023 – Jun 2023

Google: Data Analytics Specialization

Apr 2021 – Jun 2021

University of Amsterdam: Basic Statistics

Jun 2021 – Jul 2021

Academic Service

- **Reviewer**, Biomedical Signal Processing and Control (Q1 journal; 5 reviews).

Extracurricular Activities

English & Bangla News Presenter at Bangladesh Betar, Cox's Bazar, Bangladesh

Apr 2025 – Present

Member at Partibartan, Cultural Organization, Comilla University, Cumilla, Bangladesh

Jan 2014 – Jan 2017